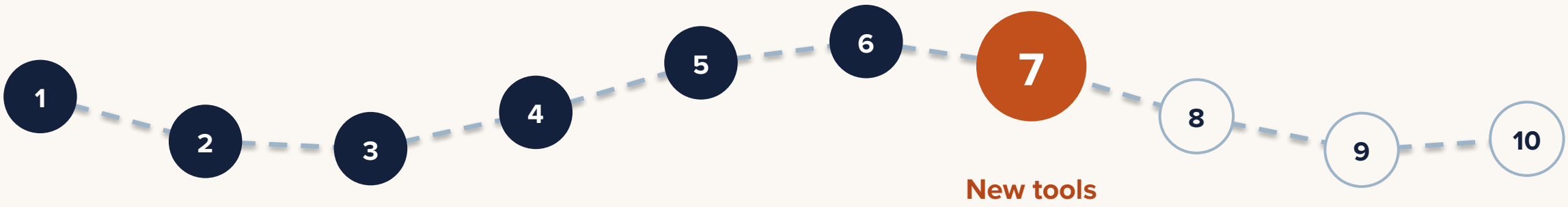
PYTHON FOR SUPPLY CHAIN

Session 7 · Mon 7 Dec 2026

New tools

SQL, Power BI, and your project question



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Today

1

SQL

`SELECT  WHERE  GROUP BY`

2

Power BI

`a look, then at home`

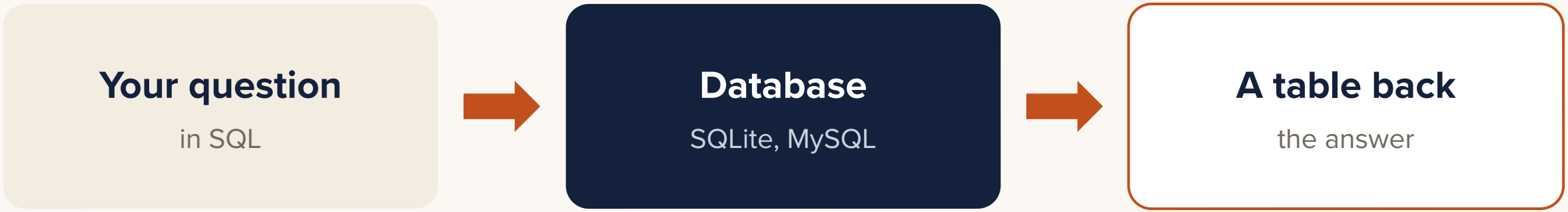
3

Your question

`project_plan.md`

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SQL asks a database a question




```
sql_table.to_sql("sales", connection, index=False)
```

SQLite lives inside Python. Nothing to install.

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Read a query line by line

`SELECT country, SUM(revenue) AS total_revenue`	which columns
`FROM sales`	which table
`GROUP BY country`	the piles
`ORDER BY total_revenue DESC`	biggest first
`LIMIT 5`	five rows

One query, one question.

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SQL and pandas: the same idea

SQL		pandas
`WHERE`	=	`df[mask]`
`GROUP BY`	=	`.groupby()`
`SUM()`	=	`.sum()`
`ORDER BY ... DESC`	=	`.sort_values(ascending=False)`
`LIMIT 5`	=	`.head(5)`

Learn the idea once.

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WHERE is the filter

```
SELECT SUM(revenue) AS germany_revenue
FROM sales
WHERE country = 'Germany'
```

```
→ 37365.63
```

`'Germany'`

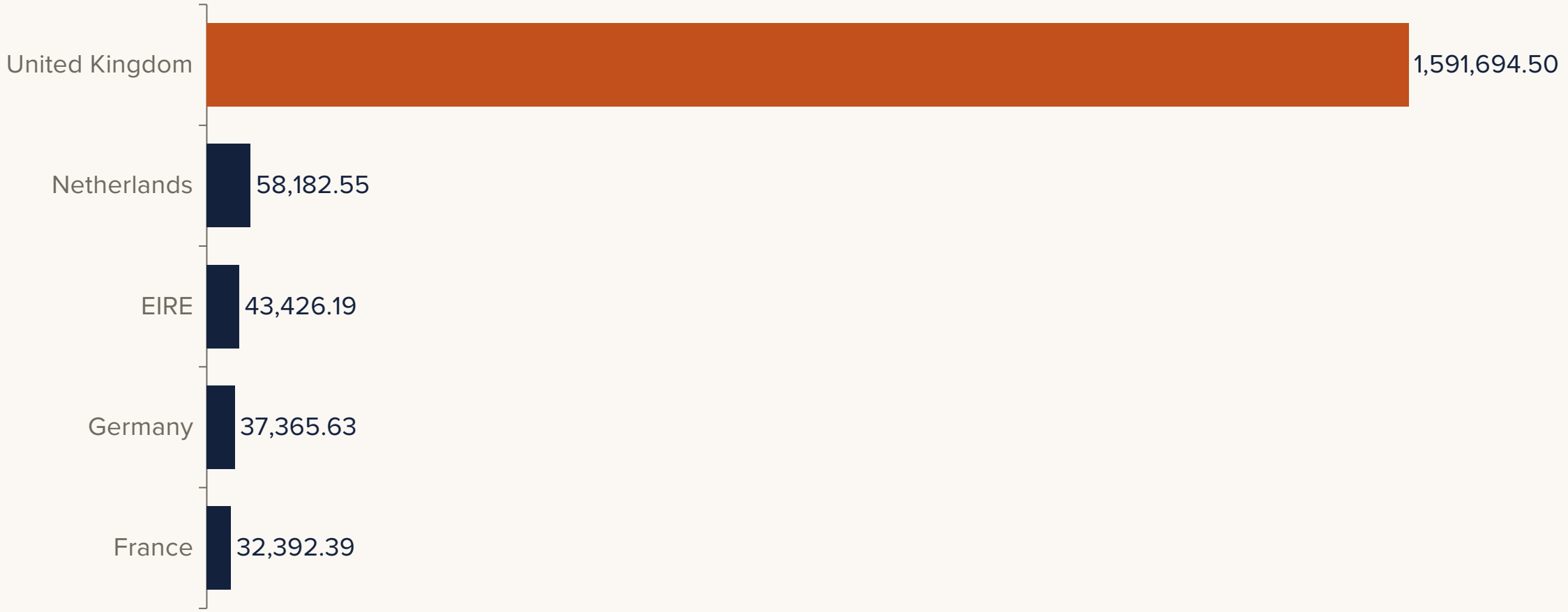
single quotes for text

`=`

one sign asks in SQL

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The first query's answer



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Power BI: a dashboard in four steps

1

Get data

Text/CSV:
product_value.csv

2

Bar chart

product and revenue

3

Card

total revenue

4

Save

send a screenshot

Watch today. Build it at home.

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A good question is...

Specific

one thing

Answerable

with this data

Useful

someone can decide

Change the question, not the skill.

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What this data can and cannot answer

We have	We do not have
stock_code, description	profit
country	stock levels
invoice_date, month	lead times
quantity, price, revenue	more than 3 months

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The project menu

1

Steady stars?

Is class A revenue steady month to month?

2

Best export market

Top country outside the UK, and its top 5 products

3

Cancellations

How big, and which products? cancellations.csv

4

Reorder point

Top 5 products, your own lead time. Label every assumption.

Or write your own.

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Your project plan

1. My question
2. Columns I need
3. Steps
4. The chart I expect
5. The decision it helps

project_plan.md

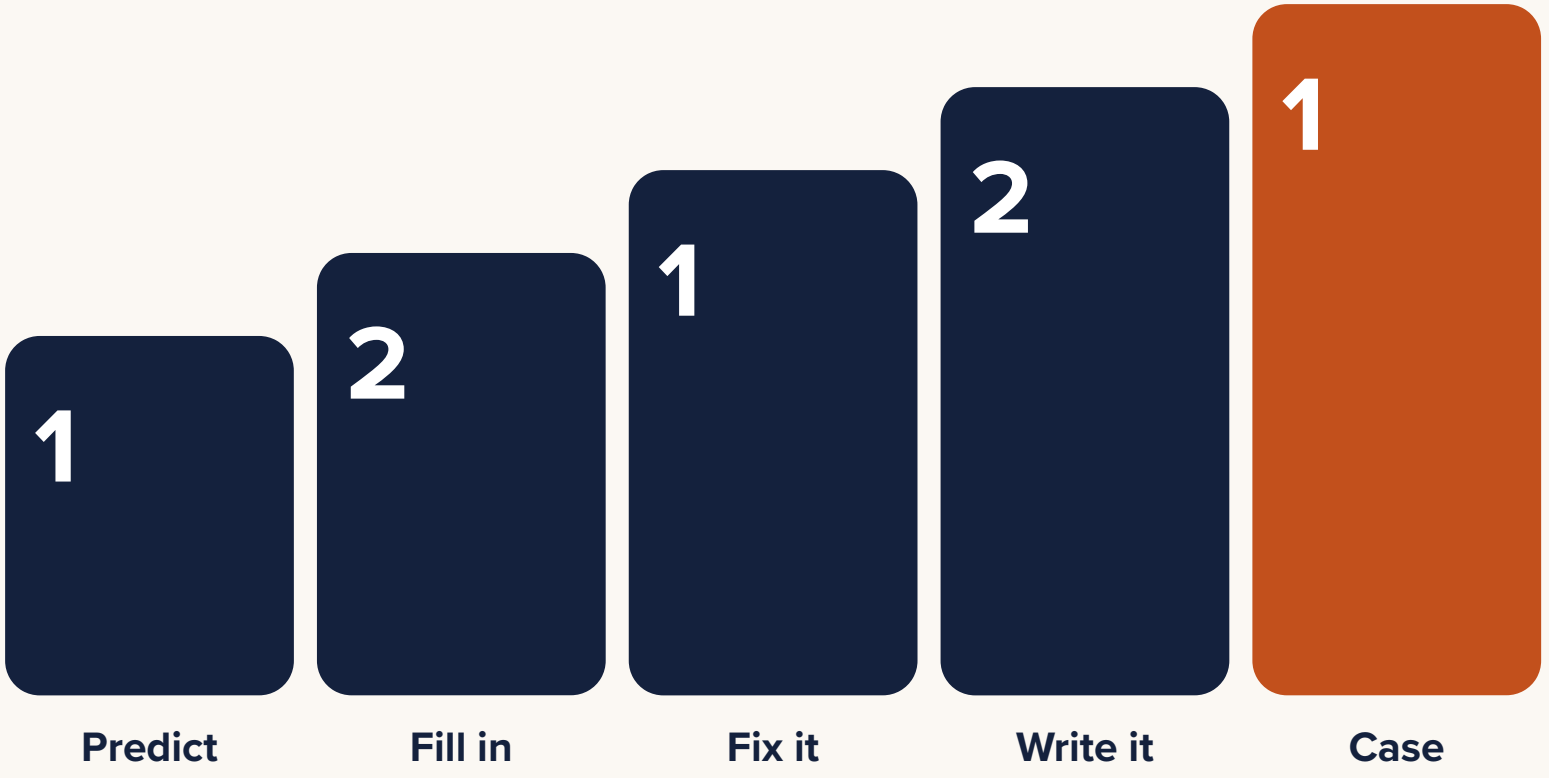
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Your SQL toolkit

SQL	Does
`SELECT col1, col2`	choose columns
`FROM sales`	the table
`WHERE country = 'Germany'`	keep some rows
`GROUP BY country`	piles
`SUM(revenue) AS total`	add and name it
`ORDER BY total DESC`	biggest first
`LIMIT 10`	the first rows
`pd.read_sql_query(query, connection)`	run it from Python

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Now in Colab



Notebook

s7_new_tools

Homework

Power BI, project plan

Recall

Tue 8, Wed 9 Dec

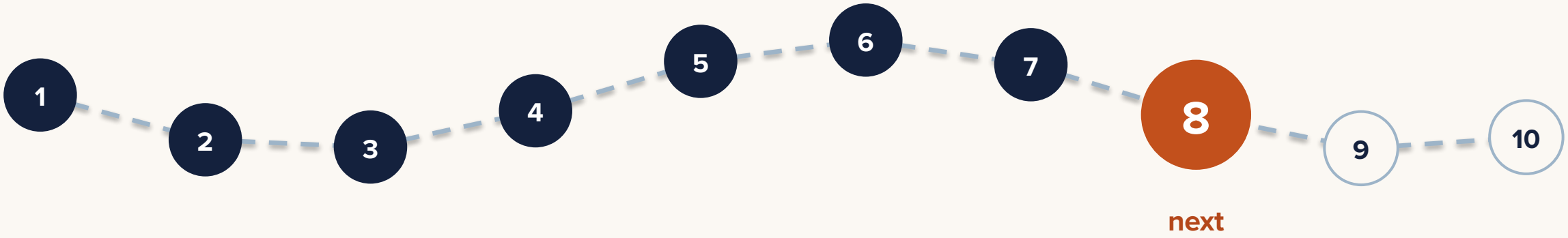
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Thank you

Next: Session 8

Detective work on your question

Thu 10 Dec



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